

BEYOND THE BASE: EXPONENT ALLOCATION IN ROPE

Anonymous authors

Paper under double-blind review

ABSTRACT

RoPE typically pairs a tunable base with an equally spaced exponent schedule. We study how reshaping that schedule changes positional representation and length generalization. A phase-invariant analysis of the full sine-cosine basis reveals how nearby slow frequencies converge to the same positional subspace. This motivates a convex surrogate allocation problem whose closed-form solution, EVQ-COSH, redistributes exponents without adding learned positional parameters. In a three-seed fixed-range training experiment, changing only the interior exponents improves every evaluated out-of-distribution length in every seed. The construction also improves long-context modeling with scarce rotary channels and under matched adaptation: at 32K, Llama-3-8B perplexity falls from 991.5 to 127.9. We then study exponent adjustments in frozen mature models, expressing frequency blends and mixed-radix transformations in a common displacement coordinate. On OLMo, an intermediate-band redistribution improves mean token F1 from 21.62% to 25.44% across five natural-QA tasks. The two stages provide a geometric account, explicit constructions, and controlled model evidence for designing RoPE through its exponent distribution.

1 INTRODUCTION

Rotary position embedding (RoPE) assigns a different rotation frequency to each pair of query and key coordinates (Su et al., 2024). In the usual parameterization, $\omega_k = b^{-\phi_k}$, a single base b and an equally spaced exponent sequence ϕ_k determine the entire table. Changing the base is a well-established way to adjust the range of positional scales (Liu et al., 2024; Xu et al., 2024; Oka et al., 2026b). The exponents make another choice: how the available rotary pairs are distributed within that range. What changes when this distribution is designed explicitly?

This question matters because each attention head has finitely many rotary pairs. Over a bounded context, several frequencies can produce almost the same positional response. For example, in a standard 64-pair table with base 500,000, the 23 pairs that rotate by at most one radian over 4K tokens have a block-whitened Rényi-2 effective rank of approximately two under uniform separations. Changing their spacing changes the basis available to the model while preserving the rotation operator. We study this allocation through its exponents, from their geometry to constructions that can be evaluated in language models.

Our first step separates exponent shape from frequency range. Figure 1 compares two tables with identical endpoints and pair count. One uses equally spaced exponents; the other redistributes the interiors using a Cosh density. Across three paired training seeds, the reallocation improves all three tested extrapolation lengths. The paired design exposes the contribution of exponent shape to length generalization.

We develop the allocation in two stages. First, we analyze the full sine-cosine subspace associated with each frequency. The analysis identifies the shared slow-frequency limit and motivates a variational problem for distributing exponents. Its solution yields EVQ-COSH, an analytic inverse-CDF construction that changes only the frequency table. We evaluate it through fixed-range controls, language-model training, and matched adaptation. With 16 rotary pairs in a 432M MLA model, it reduces 16K perplexity by 31.1%. Under matched Llama-3-8B adaptation, 32K perplexity falls from 991.5 to 127.9, with the full original-window and extrapolation curves reported together.

Second, we study how exponents can be adjusted in an already trained model. Such a model has learned to use its original table, so the adjustment is defined relative to that table. Existing

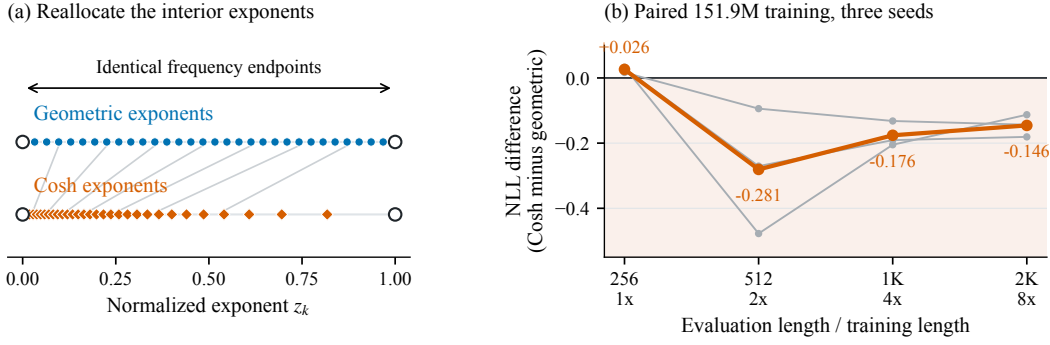


Figure 1: **Exponent allocation changes length generalization at fixed frequency range.** Left: two allocations of the same 32 rotary pairs, with identical endpoints. Right: paired 151.9M models trained at length 256; changing the 30 interior exponents lowers extrapolation NLL in every seed. Thin lines show the three training seeds and the heavy line their mean.

methods provide useful constructions: YaRN blends frequencies across bands, MrRoPE distributes a cumulative scale through mixed-radix factors, and learned-frequency methods optimize the entries (Peng et al., 2024; Tian et al., 2026; Karypis et al., 2026; Wang et al., 2026). We express these operations as exponent displacements and use matched experiments to compare their shapes. This stage includes a single static profile that improves Qwen’s RULER score over YaRN at 64K while retaining its 32K score, and a boundary-matched redistribution with gains on five OLMo natural-QA tasks.

Contributions. We (i) formulate exponent allocation independently of sampled range and analyze its phase-invariant positional geometry; (ii) derive a closed-form allocation and establish its effects through paired controls, training, and adaptation; and (iii) study structured exponent adjustments in frozen models, connecting existing transformation families to matched retrieval and natural-QA results. The common object is the exponent distribution: the two stages examine how to construct it before learning and how to adjust it after weights have been learned.

2 EXPONENT ALLOCATION

Let $K = d_{\text{rot}}/2 \geq 2$ be the number of rotary pairs and $b > 1$ the base. Standard RoPE uses

$$\omega_k = b^{-\phi_k}, \quad \phi_k = \frac{k}{K}, \quad k = 0, \dots, K-1. \quad (1)$$

The base controls the spacing of this geometric frequency family. Allowing ϕ_k to follow a non-linear sequence also changes the density of samples across positional scales. To describe that shape without conflating it with an overall shift or stretch, take an ordered table $\omega_0 > \dots > \omega_{K-1} > 0$ and write

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (2)$$

Here $a = x_0$ and $R = x_{K-1} - x_0$ specify the sampled log-frequency range. Geometric tables have $z_k = k/(K-1)$ at every base. Holding (a, R) fixed while changing z therefore changes only the normalized exponent allocation.

This representation gives both a construction and an experimental control. A density over exponents generates a finite table through its quantiles. The same quantiles can either be installed directly or affinely mapped onto fixed endpoints to test the effect of their shape. We use the anchored version for fixed-range comparisons and state the installed grid in each model protocol. For a mature checkpoint, the corresponding variable is the displacement from its original exponents; §5 develops that form.

The analysis below asks what positions these finitely many exponents can represent. The experiments then ask how models learn to use the resulting basis and how existing weights respond to a change in it.

3 GEOMETRY AND A CLOSED-FORM ALLOCATION

3.1 WHAT DIFFERENT EXPONENTS REPRESENT

A rotary pair contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is the relative position and C, D depend on query and key content. The positional object associated with its exponent is therefore the subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

Comparing these two-dimensional spaces accounts for every content phase within a pair. Let $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$, and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a specified distribution of separations. For nonsingular S_ω, S_ν , the singular values of $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations of the two spaces. Their average squared correlation,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

is unchanged by rotation of the phase inside either pair. It measures how much of the positional basis one frequency shares with another.

Stacking all block-whitened pairs gives a Gram matrix Γ with I_2 on its diagonal blocks. If $\bar{c}$ is the mean of c_{ij} over ordered distinct pairs, its Rényi-2 effective rank satisfies

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

At fixed K and range, exponent allocation changes this basis through the off-diagonal correlations. For uniform separations on $[0, L]$, the cross-Gram has an exact trigonometric form, so both the correlations and effective rank can be computed directly (App. A.1).

3.2 WHY THE SLOW END ADMITS REALLOCATION

Proposition 1 (Shared slow-frequency subspace). *Under the uniform measure on $[0, L]$, as $\omega L \rightarrow 0$, V_ω approaches $\text{span}\{1, \Delta\}$. If $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, then $2 - \|Q_{\omega\nu}\|_F^2 = O(\epsilon^4)$.*

The limit follows from $\cos(\omega\Delta) \rightarrow 1$ and $\sin(\omega\Delta)/\omega \rightarrow \Delta$; Appendix A.1 derives the leading fourth-order coefficient. Several slow pairs can thus supply many coordinate dimensions but very few distinct positional directions. For $b = 500,000$, $K = 64$, and $L = 4096$, the 23 standard-grid pairs with $\omega L \leq 1$ have 46 nominal dimensions and $r_2 = 2.00013$ under this measure.

This suggests redistributing some exponent samples toward faster rotations while retaining coverage of the range. Frequency-use studies also show that slow pairs can carry semantic information (Barbero et al., 2025; Urrutia et al., 2026); their positional similarity motivates a change in allocation, not deletion of their content coordinates. The learned weights determine how the resulting basis is used.

3.3 CONSTRUCTING AN EXPONENT DISTRIBUTION

Let a positive density $\rho \in C^2([0, 1])$, with $\int_0^1 \rho = 1$, allocate samples over the exponent interval. Uniform ρ recovers geometric spacing. To balance concentration against redundant slow-end sampling, we choose the convex objective

$$C_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad \alpha > 0, \beta \geq 0. \quad (6)$$

The squared-density term discourages concentration in a narrow band. The interaction penalty grows when both exponents lie toward the slow end. This surrogate turns the preceding geometric observation into a tractable allocation problem.

Theorem 2 (Cosh allocation). *The unique minimizer of (6) over positive $\rho \in C^2([0, 1])$ with unit integral is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1 - \phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (7)$$

with $\rho_0(\phi) = 1$ obtained by continuity at $\beta = 0$.

The Euler–Lagrange equation reduces to $\rho'' = \tau^2 \rho$ with $\rho'(1) = 0$; normalization fixes its scale. Integrating and inverting the CDF gives the finite table

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k + 1/2}{K}. \quad (8)$$

We call this construction EVQ-COSH. For $\tau > 0$, its density decreases from the fast to the slow end, allocating more samples to faster positional variations. The limit $\tau \rightarrow 0$ gives uniform-log midpoint quantiles.

Installation requires evaluating (8) once and setting $\omega_k = b^{-\phi_k}$. The frequency table stays fixed during training or adaptation, and the attention operator and trainable parameter count are unchanged. For the fixed-range experiment we normalize these quantiles using (2) and anchor them to the baseline endpoints. The grid conventions, values of τ , and operating-rule studies are given in Apps. A.9 and C.

4 LEARNING WITH DESIGNED EXPONENTS

We first test exponent allocation with weights trained or adapted to the installed table. The paired fixed-range experiment isolates its shape; the larger-model studies measure language modeling and retrieval under their respective training protocols.

4.1 DOES EXPONENT SHAPE MATTER AT FIXED RANGE?

We trained paired 151.9M models at length 256 with $K = 32$ rotary pairs. Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The geometric arm follows the FMROPE training-base rule $b = L_{\text{train}}$ (Oka et al., 2026b); the Cosh arm anchors its $\tau = 4$ quantiles to the same two endpoints. Only the 30 interior exponents differ.

At the fixed training range, Cosh lowers tail NLL at 512, 1K, and 2K in all three seeds (Fig. 1). The mean Cosh-minus-geometric differences at 256/512/1024/2048 are $+0.026 / -0.281 / -0.176 / -0.146$. Exponent shape therefore changes learned length generalization even when no scalar base change is available to explain the result.

A separate evaluation retargets each table’s range to the declared test length while preserving its normalized exponent shape. This intervention favors the geometric arm, with Cosh-minus-geometric NLL $+0.060 / +0.227 / +0.460$ at 512/1K/2K. Range selection and exponent allocation consequently need to be evaluated together when composing a deployment table. The complete paired curves and training details appear in Appendix C.

A further 50.9M factorial varies two bases, two training lengths, three head dimensions, and three paired seeds, with endpoints pinned in every comparison. It evaluates three Cosh strengths and a deformation-matched exponential. The strongest tested Cosh setting improves the weighted OOD objective in 10/12 configurations, and the exponential in 9/12 (App. Table 13). The common benefit across these structured shapes extends the fixed-range result beyond a single curve.

4.2 LANGUAGE MODELING AND RETRIEVAL

Scarce rotary channels. The 432M MLA model uses 16 rotary pairs and trains on 500M tokens at length 8192 (DeepSeek-AI, 2024a;b). Across three seeds, EVQ-COSH reduces 16K perplexity from 138.8 to 95.6; at 8K the corresponding values are 35.4 and 35.8 (Table 1). This tests exponent allocation in an architecture where only a small set of dimensions carries explicit rotation.

Continued full-parameter training. In the matched 750M continuation, both arms train at length 4096. At 16K, EVQ-COSH reduces perplexity from 45.1 to 24.4, while the 4K values remain close at 22.0 and 22.3. Across 40 associated 8K passkey trials, greedy autoregressive exact match rises from 0 to 77.5%. The per-length results and generation protocol are in Appendix B.

Matched Llama-3-8B adaptation. We compare Native and EVQ-COSH under the same 300-step rank-64 Q/K/V/O LoRA adaptation, using physical 8K sequences (Hu et al., 2022; Grattafiori et al.,

Table 1: **Exponent allocation in trained language models.** Each row compares Geo and EVQ-COSH within one training protocol. Perplexity is lower-is-better; the MLA values aggregate three seeds, and the continuation uses one matched pair.

Model	Training length	Evaluation	Geo	EVQ-COSH
432M MLA	8K	8K	35.4	35.8
		16K	138.8	95.6
750M continuation	4K	4K	22.0	22.3
		16K	45.1	24.4

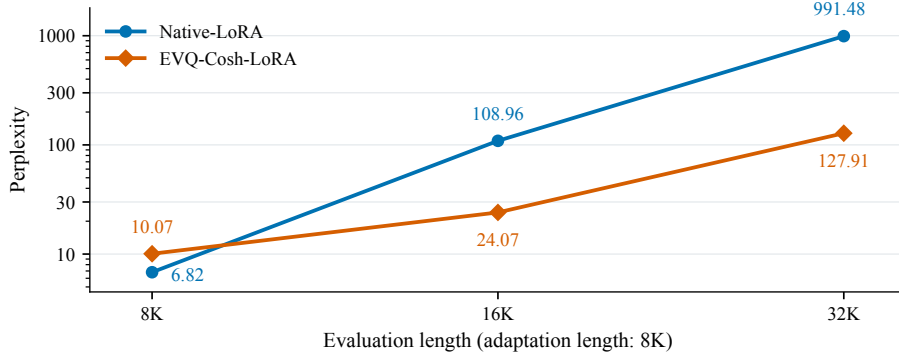


Figure 2: **The effect of exponent allocation grows beyond the adaptation window.** Llama-3-8B arms share an 8K, 300-step LoRA protocol and are evaluated on the same 24 text packs. The logarithmic PPL axis shows all three evaluation lengths, including the original window.

2024). The evaluation comprises 24 frozen temporal text packs from three domains. Figure 2 shows the full curve: Native-LoRA/EVQ-COSH-LoRA PPL is 6.82/10.07 at 8K, 108.96/24.07 at 16K, and 991.48/127.91 at 32K. The long-context NLL improvement holds in every pack in both extended lengths.

Source interventions help interpret this result. On ten true-16K passkey cases, median target-block hit@16 rises from 18.75% to 64.06%. Removing decode attention to the remote gold block raises answer NLL by 1.5055 in the EVQ-COSH arm, compared with -0.0095 in Native. Thus the adapted model’s answer probability depends on the distant source. Appendix E.3 reports this intervention and the separate RULER-family continuation.

Further settings. From-initialization OLMo results and video-DiT experiments are collected in Apps. E and B.

5 ADJUSTING EXPONENTS IN FROZEN MODELS

A model trained with one exponent distribution has learned query and key coefficients in that basis. The weights-by-table crossing makes this dependence visible: in two 50M models, replacing the Geo-trained model’s table by Cosh changes PPL from 7.14 to 76.20; the reverse swap changes 7.16 to 23.05 (App. Table 5). For frozen-model extension, we therefore study adjustments relative to the native table.

5.1 A COMMON COORDINATE FOR EXPONENT ADJUSTMENTS

For native frequencies ω_k^N and an installed table ω'_k , define

$$d_k = \log \frac{\omega_k^N}{\omega'_k}, \quad \omega'_k = \omega_k^N e^{-d_k}, \quad \phi'_k - \phi_k^N = \frac{d_k}{\log b}. \quad (9)$$

Table 2: **Frozen-model RULER scores (%) at matched exponent range.** OLMo uses nine tasks at 16K; Qwen uses four at 64K, with 20 rows per task. Within each model only the interior allocation changes.

Exponent allocation	OLMo-1B	Qwen-1.5B
Geometric	0.56	57.75
Coarse ramp	61.04	64.00
Residual-guided	60.47	66.50

Table 3: **A static extension across the native and extended windows.** Qwen-0.5B RULER-13 macro (%); g is the fixed cosine/sine amplitude used by each method.

Installed table	g	32K	64K
Native	1.0000	54.78	22.05
YaRN-2	1.0693	55.94	45.37
Normalized-index-2	1.0513	55.92	51.46

Several familiar constructions then have explicit displacement profiles:

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (10)$$

Here s is the extension factor, w_k is a blend weight, and m_k is the normalized cumulative exponent shift. YaRN’s NTK-by-parts frequency map uses the first construction; MrRoPE uses the third (Peng et al., 2024; Tian et al., 2026). The coordinate lets us compare their realized exponent shapes directly.

5.2 FIXED-RANGE AND SINGLE-TABLE COMPARISONS

Changing only the normalized exponent allocation. On frozen OLMo-2-1B and Qwen-2.5-1.5B, we hold the endpoints, 64 pairs, attention amplitude, inputs, and decoder fixed. A geometric allocation, a residual-guided allocation, and its nearest coarse ramp give the results in Table 2. The residual-guided construction moves pairs according to their remaining energy after projection onto the other sine–cosine pairs; the ramp approximates that profile without task labels. Both structured profiles produce large gains over uniform exponents. Their definitions and reference Native/YaRN rows are in Appendix E.2.

One static table at both lengths. A frozen normalized-index profile transfers a reference displacement curve onto the native slot grid and installs $\omega'_k = \omega_k^N s^{-m_k}$. On Qwen-2.5-0.5B, a new-seed 13-task RULER evaluation uses 20 examples per task at 32K and 64K. Every request uses the same $s = 2$ table throughout prefill and decoding, including requests within the native window. Table 3 shows a 6.09-point advantage over YaRN at 64K, with paired task-stratified bootstrap interval $[2.76, 9.58]$ points. At 32K the two methods are nearly equal. Appendix F.1 gives the frozen profile and the full task breakdown.

5.3 REDISTRIBUTING THE INTERMEDIATE-BAND SHIFT

MrRoPE-Pro progressively distributes the cumulative shift across an intermediate frequency band. We compare it with a boundary-matched (BM) profile, which keeps the same band endpoints and total scale while smoothing the per-slot radix increments toward the unmodified outer bands. Both are explicit constructions of m_k in (10); their exact discrete formulas are given in Appendix F.2.

In an OLMo-1B six-task confirmation, BM improves the 16K macro from 2.78% for MrRoPE-Pro to 51.32%; MrRoPE-Uni scores 32.12%. The comparison extends to natural inputs: across 631 eligible long examples from five QA tasks, the task-equal F1 rises from 21.62% to 25.44%. The paired interval for the improvement is $[1.32, 6.29]$ points. Every task has a positive mean difference (Fig. 3). All methods use a fixed factor-four table; the input and decoding details are in Appendix F.4.

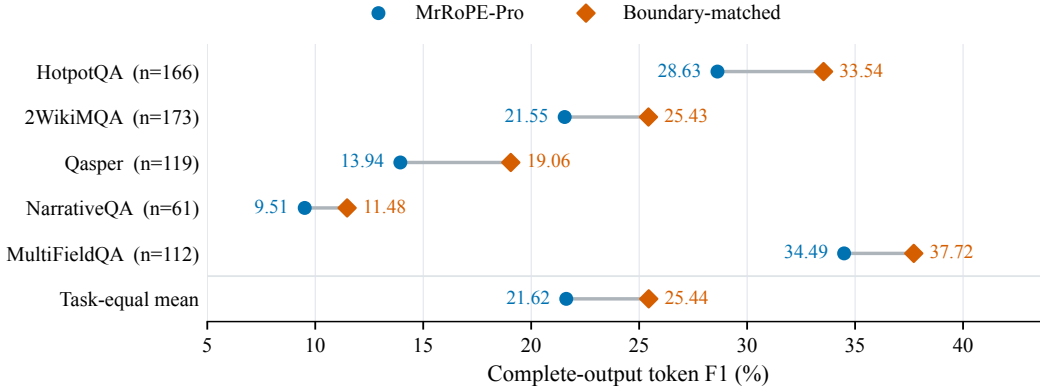


Figure 3: **An intermediate-band redistribution improves OLMo natural QA.** Complete-output token F1 on untruncated long inputs, using the same $s = 4$ configuration and decoder in both arms. Sample counts are shown by task; the final row averages the five task means.

Table 4: **Matched six-task RULER comparisons (%)**. Each length column averages six task scores. Both methods use static $s = 4$; the row counts refer to short/long prompts per arm.

Model	Lengths	Rows	Short		Long	
			MrPro	BM	MrPro	BM
OLMo-1B	4/16K	24/48	37.85	81.81	2.78	51.32
Qwen-3B	32/128K	12/24	87.22	91.67	78.13	70.83
Qwen-7B	32/128K	6/12	83.33	80.00	84.44	71.11

The same construction gives a different outcome on Qwen (Table 4). In the six-task Qwen-3B comparison, BM improves the 32K mean but MrRoPE-Pro is stronger at 128K. On the smaller Qwen-7B screen, MrRoPE-Pro leads at both lengths. These comparisons show why exponent shape must be evaluated with the checkpoint and target length, even when the band boundaries and cumulative scale agree. The full per-task results are retained in Appendix F.3.

6 RELATED WORK

Range, scaling, and exponent shape. RoPE’s rotation operator (Su et al., 2024) admits many frequency tables. Scaling-law and base-bound analyses study how the geometric family’s range affects length generalization (Liu et al., 2024; Xu et al., 2024). FMRoPE relates the learned frequency band to the base and training length, and chooses a base accordingly (Oka et al., 2026b). In that family the normalized exponents remain equally spaced; our fixed-range comparisons vary their interior distribution.

Position interpolation, YaRN, and LongRoPE extend contexts through uniform or frequency-dependent transformations (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025). LongRoPE combines dimension-wise scaling with a token-position threshold; the dimension-wise factors correspond to exponent displacements. MrRoPE makes the cumulative scaling structure explicit through mixed-radix conversion and gives uniform and progressive intermediate-band rules (Tian et al., 2026). Equation (10) relates those factors to exponent displacements. Our second-stage comparisons build on this connection; the first stage develops a distributional construction and tests exponent shape while holding the frequency endpoints fixed.

Learning and using frequency bands. LeRoPE learns one log-space scale per rotary pair, shared across layers and heads, and shows that both the resulting table and its joint training contribute to performance (Karypis et al., 2026). AdaRoPE learns head-specific frequencies and attention scaling, including comparisons with learned-base parameterizations (Wang et al., 2026). These approaches optimize the same frequency entries that can be expressed as exponent changes. We

instead analyze the allocation geometry, derive an explicit density, and use controlled interventions to separate allocation from sampled range.

Frequency-use studies connect fast and slow bands to positional and symbolic attention, retrieval, and data-dependent scales (Barbero et al., 2025; Urrutia et al., 2026; Chiang & Yogatama, 2025; Oka et al., 2026a; Wu et al., 2026). DoPE uses truncated matrix entropy to identify low-rank structures in rotated activations and selectively modifies positional encoding (Xiong et al., 2025). Our canonical correlations compare entire sine-cosine subspaces as the exponents vary. This separates a property of the supplied positional basis from the frequency usage learned by a particular model.

Changing the positional operator. FoPE introduces Fourier components, while HoPE and clipped encodings alter the treatment of slow rotations (Hua et al., 2025; Chen et al., 2025; Li et al., 2026a). GRAPE generalizes positional transformations through group actions; Selective RoPE learns input-dependent rotations (Zhang et al., 2026; Movahedi et al., 2026). RePo learns contextual token positions before applying the encoding (Li et al., 2026b). These works change the positional map or its dependence on content. Our construction retains the standard rotation and designs its exponent table. Recent analyses of long-context positional and token confusion further motivate studying the interaction between frequency choice and the representations it acts on (Du et al., 2026).

7 DISCUSSION

Exponent allocation connects a simple implementation choice to the positional basis a model learns to use. The fixed-range experiments show that reshaping the exponents changes length generalization; the subspace analysis explains how that reshaping changes the distribution of positional directions. EVQ-COSH turns this view into a closed-form construction, with substantial long-context gains across the trained and adapted models.

The frozen-model results add a second perspective. Expressing a change as a displacement from the native exponents makes blends, log-frequency shifts, and mixed-radix rules directly comparable. Their different outcomes across checkpoints show that the original learned representation is part of the design problem. An exponent table should therefore be specified together with its sampled range and the way it is installed.

A natural next step is to study exponent allocation under sparse attention. Selective access to tokens and compressed representations change which positional distinctions reach the reader. The framework developed here provides a way to ask how the exponent distribution should serve those architectures.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. Figure sources and numerical summaries are bound to recorded experiment artifacts, and the supplementary implementation provides the closed-form construction and exponent-adjustment formulas. Training, evaluation, and analysis records are kept with their corresponding protocols so that the reported tables and figures can be reconstructed.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

REFERENCES

- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3119–3137, 2024. doi: 10.18653/v1/2024.acl-long.172. URL <https://aclanthology.org/2024.acl-long.172/>.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html.
- Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usvsn Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*, pp. 95–136, 2022. doi: 10.18653/v1/2022.bigscience-1.9. URL <https://aclanthology.org/2022.bigscience-1.9/>.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023. doi: 10.48550/arXiv.2306.15595. URL <https://arxiv.org/abs/2306.15595>.
- Yuhan Chen, Ang Lv, Jian Luan, Bin Wang, and Wei Liu. HoPE: A novel positional encoding without long-term decay for enhanced context awareness and extrapolation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 23044–23056, 2025. doi: 10.18653/v1/2025.acl-long.1123. URL <https://aclanthology.org/2025.acl-long.1123/>.
- Ting-Rui Chiang and Dani Yogatama. The rotary position embedding may cause dimension inefficiency in attention heads for long-distance retrieval. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 13552–13562, 2025. doi: 10.18653/v1/2025.findings-acl.697. URL <https://aclanthology.org/2025.findings-acl.697/>.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a. doi: 10.48550/arXiv.2405.04434. URL <https://arxiv.org/abs/2405.04434>.
- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.

- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pp. 11091–11104, 2024. URL <https://proceedings.mlr.press/v235/ding24i.html>.
- Yufeng Du, Phillip Harris, Minyang Tian, Eliu A. Huerta, Srikanth Ronanki, Subendhu Rongali, Aram Galstyan, and Hao Peng. RoPE distinguishes neither positions nor tokens in long contexts, provably. *arXiv preprint arXiv:2605.15514*, 2026. doi: 10.48550/arXiv.2605.15514. URL <https://arxiv.org/abs/2605.15514>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783. URL <https://arxiv.org/abs/2407.21783>.
- Robert M. Gray and David L. Neuhoff. Quantization. *IEEE Transactions on Information Theory*, 44(6):2325–2383, 1998. doi: 10.1109/18.720541.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580. URL <https://aclanthology.org/2020.coling-main.580/>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Ermo Hua, Che Jiang, Xingtai Lv, Kaiyan Zhang, Youbang Sun, Yuchen Fan, Xuekai Zhu, Biqing Qi, Ning Ding, and Bowen Zhou. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 24932–24949, 2025. URL <https://proceedings.mlr.press/v267/hua25b.html>.
- Petros Karypis, Sean O’Brien, Shreyas Kadekodi, Rui Zhu, and Julian McAuley. LeRoPE: Learnable RoPE frequencies improve language modeling. *arXiv preprint arXiv:2607.10134*, 2026. doi: 10.48550/arXiv.2607.10134. URL <https://arxiv.org/abs/2607.10134>.
- Haoran Li, Sucheng Ren, Alan Yuille, and Feng Wang. CoPE: Clipped RoPE as a scalable free lunch for long context LLMs. *arXiv preprint arXiv:2602.05258*, 2026a. doi: 10.48550/arXiv.2602.05258. URL <https://arxiv.org/abs/2602.05258>.
- Huayang Li, Tianyu Zhao, Deng Cai, and Richard Sproat. RePo: Language models with context re-positioning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026b. URL <https://icml.cc/virtual/2026/poster/64002>. arXiv:2512.14391.
- Xiaoran Liu, Hang Yan, Chenxin An, Xipeng Qiu, and Dahua Lin. Scaling laws of RoPE-based extrapolation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=JO7k0SJ5V6>. arXiv:2310.05209.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017. URL <https://openreview.net/forum?id=Byj72udxe>. arXiv:1609.07843.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=AQo1SEElNb>.
- Yui Oka, Kentaro Hanafusa, Taku Hasegawa, Kyosuke Nishida, and Kuniko Saito. Probing rotary position embeddings through frequency entropy. In *International Conference on Learning Representations (ICLR)*, 2026a. URL <https://openreview.net/forum?id=1JZuEDq62N>.

- Yui Oka, Itsumi Saito, Kyosuke Nishida, and Kuniko Saito. Frequency bands in RoPE: Base frequency and context length shape the interpolation–extrapolation trade-off. In *International Conference on Learning Representations (ICLR)*, 2026b. URL <https://openreview.net/forum?id=PR1PPxvG9Q>.
- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024. doi: 10.48550/arXiv.2406.17557. URL <https://arxiv.org/abs/2406.17557>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=wHBfxhZulu>.
- Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 54203–54218, 2025. URL <https://proceedings.mlr.press/v267/shang25a.html>.
- Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pp. 843–852, 2015. URL <https://proceedings.mlr.press/v37/srivastava15.html>.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025. doi: 10.48550/arXiv.2501.00656. URL <https://arxiv.org/abs/2501.00656>.
- Qingyuan Tian, Wenhong Zhu, Xiaoran Liu, Xiaofeng Wang, and Rui Wang. MrRoPE: Mixed-radix rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/69413f87e5a34897cd010ca698097d0a-Abstract-Conference.html.
- Felipe Urrutia, Jorge Salas, Alexander Kozachinskiy, Cristian Buc Calderon, Hector Pasten, and Cristobal Rojas. Decoupling positional and symbolic attention behavior in transformers. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=V38yAoqddQ>.
- Shaowen Wang, Yuke Zheng, Tansheng Zhu, Shuang Chen, Shaofan Liu, Suncong Zheng, and Jian Li. AdaRoPE: Not all attention heads should rotate and scale equally. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/60704>. arXiv:2607.19363.
- Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for good video rotary position embedding? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 66118–66136, 2025. URL <https://proceedings.mlr.press/v267/wei25h.html>.
- Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. How data shapes RoPE frequency usage: From positional scale matching to length generalization. *arXiv preprint arXiv:2607.07678*, 2026. doi: 10.48550/arXiv.2607.07678. URL <https://arxiv.org/abs/2607.07678>.

- Jing Xiong, Liyang Fan, Hui Shen, Zunhai Su, Min Yang, Lingpeng Kong, and Ngai Wong. DoPE: Denoising rotary position embedding. *arXiv preprint arXiv:2511.09146*, 2025. doi: 10.48550/arXiv.2511.09146. URL <https://arxiv.org/abs/2511.09146>.
- Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2773. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/9f12dd32d552f3ad9eaa0e9dfec291be-Abstract-Conference.html.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. doi: 10.48550/arXiv.2412.15115. URL <https://arxiv.org/abs/2412.15115>.
- Yifan Zhang, Zixiang Chen, Yifeng Liu, Zhen Qin, Huizhuo Yuan, Kangping Xu, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Group representational position encoding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf.
- Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 77539–77557, 2025. URL <https://proceedings.mlr.press/v267/zhao25m.html>.
- Chuanyang Zheng, Yihang Gao, Han Shi, Minbin Huang, Jingyao Li, Jing Xiong, Xiaozhe Ren, Michael K. Ng, Xin Jiang, Zhenguo Li, and Yu Li. DAPE: Data-adaptive positional encoding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0838. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/2f050fa9f0d898e3f265d515f50ae8f9-Abstract-Conference.html.

A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (11)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. The metric is consequently attached to the positional subspaces themselves, not to a chosen sine/cosine phase convention.

Theorem 3 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K - 1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 3. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K - 1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (12)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (13)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 , rather than comparing the unwhitened Taylor coefficients. Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (14)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (15)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (16)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1} M) + O(\epsilon^6) = \frac{19}{12600} (x^2 - y^2)^2 + O(\epsilon^6), \quad (17)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \rightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (18)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. This is a limit for a declared fixed measure; it does not identify which slow channels a trained model uses. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram. The fixed-range trained-model contrast of §4.1 instead follows its paired empirical protocol.

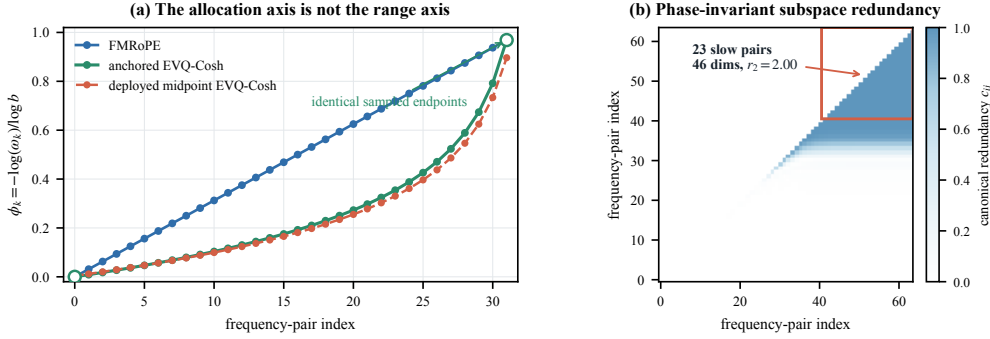


Figure 4: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

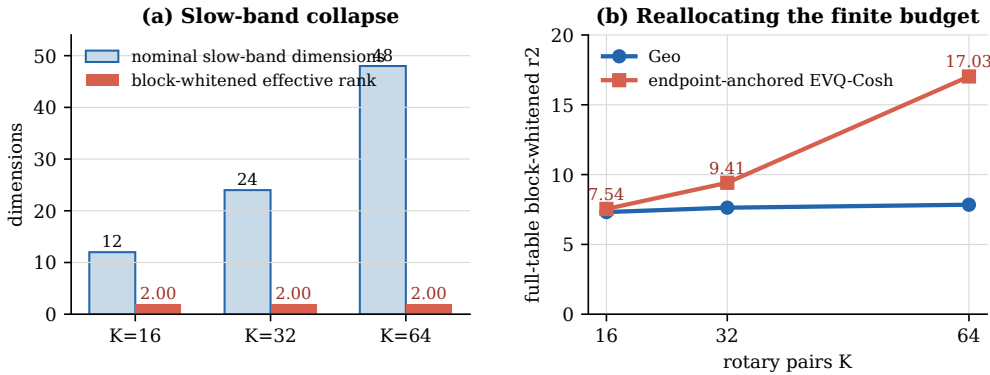


Figure 5: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows. These are static positional-basis diagnostics; the trained effect is identified separately in App. C.

A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 4 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (19)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (20)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (21)$$

so every entry of the cross-Gram in (11) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_\star((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (20) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k\Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 4 attains the algebraic rank maximum on the declared interval while imposing an exact recurrence just outside it. It does not say that K_{par} is a global capacity over arbitrary frequencies, nor does it survive a change of measure: the causal separation histogram is triangular rather than uniform and does not make the same lattice orthogonal. The exact parity construction and the small-model checks are reproduced by `analysis/full_rope_audit/verify_small_models.py`; the uniform-versus-causal measure comparison is exposed by `scripts/analysis/third_axis_ceiling.py-decompose`. These are static basis statements, not LM-quality or extrapolation predictions.

A.3 TWO STATIC ORDERING COUNTEREXAMPLES

Even within a fixed declared measure, a cheaper static proxy need not order tables like the full basis, and an ordering at one interval need not persist at another. The following two finite tables are existence counterexamples; no claim of continuous global optimisation is needed.

Cosine-only collision can choose the lower-rank table. Use $L=4096$, $b=500,000$, $K=64$, continuous uniform $\Delta \in [0, L]$, and pin both physical endpoints to $[2 \times 10^{-6}, 1]$. Let A be the best-found schedule returned by the declared heuristic for cosine-only collision and B the corresponding best-found schedule for full-subspace collision, using their distinct harmonic-augmented candidate pools. The cosine proxy prefers A ,

$$C_{\text{cos}}(A) = 2.62 \times 10^{-9} < 1.44 \times 10^{-8} = C_{\text{cos}}(B), \quad (22)$$

but the phase-invariant block-whitened basis has the opposite order,

$$r_2(A) = 64.48 < 128.00 = r_2(B). \quad (23)$$

Here C_{cos} is the mean squared normalised cosine-cosine overlap. The counterexample isolates why one phase slice cannot replace the full two-plane object: a small cosine overlap can hide redundancy in the sine and cross terms.

Full-subspace collision can reverse across lengths. Under the same base, endpoints, and training-interval convention, the tracked counterexample search emits two fixed $K=16$ tables A and B . Writing C_M for their mean phase-invariant full-subspace collision when the uniform measure spans $[0, M]$, their ordering is

$$C_L(A) = 0.45837 < 0.61695 = C_L(B), \quad (24)$$

$$C_{2L}(A) = 0.45834 > 0.41132 = C_{2L}(B), \quad (25)$$

$$C_{4L}(A) = 0.45830 > 0.24763 = C_{4L}(B). \quad (26)$$

Recomputing the same schedules on discrete uniform separations $\Delta = 0, \dots, M-1$ preserves the reversal. Thus even the correct phase-invariant collision is conditional on the declared interval; minimising it at L does not supply an extrapolation ordering at $2L$ or $4L$.

Numerical owner and reproduction convention. The digits in Eqs. (23)–(26) are frozen outputs produced by `scripts/analysis/full_rope_collision_audit.py` with `-length 4096 -base 500000 -candidate-grid 1025 -random-schedules 3000`; the counterexample RNG seed is 20260819 and the JSON stores each schedule and its float64 hash. These are best-found points returned by the declared greedy-plus-exchange heuristic in metric-specific candidate pools, not certified pool or continuous optima. The first inequality is also reported with raw full entropy rank $93.44 < 120.03$; the stable-rank ordering above is the quantity governed by Theorem 3.

Together, the parity recurrence and the ordering reversals separate three questions: dimension over a declared interval, multiscale coverage outside it, and behaviour after training. The fixed-support experiments evaluate the last question, while Table 5 shows why learned coefficients must stay with the basis that trained them.

A.4 POST-HOC TRANSPLANT OBSTRUCTION

Table 5: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Theorem 5 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta) B = \mathcal{R}_\Omega(\Delta)$ on an open interval containing zero, then Ω' and Ω have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Proof of Theorem 5. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This exact statement covers *fixed, position-independent, invertible* compensation. Approximate retraining on finite data, nonlinear maps, and changes to V/O or the residual path are separate intervention classes.

A.5 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 2 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (27)$$

the main-text collision functional is (6), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (28)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (29)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (30)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (31)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (30) and (31) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1 - \phi))}{\sinh \tau}, \quad (32)$$

which is main-text Theorem 2. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1)=\tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0)=\tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (32).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (33)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The homogeneous Cosh solution has an analytically invertible CDF, turning the continuous optimum into an explicit finite table. The functional test below checks the resulting allocation against a closed-form cosine-feature slice, and the trained comparisons supply the behavioural test.

A.6 COSINE-SLICE DIAGNOSTIC FOR THE SURROGATE-DERIVED TABLE

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ supplies the closed-form density. We validate its allocation functionally against the closed-form cosine-feature kernel $K_{\text{cos}}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This tests whether the same direction holds for the declared cosine-feature slice.

Table 6 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute K_{cos} at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii}K_{jj})$ and Shannon effective rank (the exponential of the cosine-feature Gram eigenvalue distribution). The inclusive grid $u_k = k/(K-1)$ exposes the complete diagnostic interval. This slice-level static diagnostic is separate from the phase-invariant full-subspace geometry of Theorem 3; trained midpoint tables are evaluated in §4. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$.

Table 6: Cosine-slice surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases Shannon effective rank of the cosine-feature Gram under the closed-form kernel K_{cos} in all 12 tested configurations. Theorem 3 supplies the separate phase-invariant full-subspace geometry; §4 evaluates trained midpoint tables.

Configuration	Collision C		Shannon rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32, b=500K, \text{ sweep } L$</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32, L=2048, \text{ sweep } b$</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16, L=32, \text{ sweep } b$</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under K_{cos} in all 12 configurations (24%–92%), and Shannon effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.7 SINGLE-CROSSING BUDGET SHIFT

Lemma 6 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (34)$$

Proof of Lemma 6. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving

$\rho_\tau(\phi_c) = 1$ gives (34). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\text{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1 - \phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

A.8 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 7 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (35)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (36)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (37)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (35). $\square$

An immediate corollary follows.

Corollary 8 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (38)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity checks the two surrogate energy terms exactly; trained-loss evidence is supplied by §4.

A.9 ASSUMPTION-BOUND OPERATING CALCULATION AND FINITE TESTED RECEIPT

The variational theorem selects the Cosh *shape*; it does not select a finite strength for every model or protocol. The implementable reference used in the standard full-RoPE text experiments is

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2}) / K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (39)$$

Here $c=1$ is the pre-training zero-search convention, not a fitted optimum; each protocol records its realised τ , grid, support, and rotary width.

Assumption-bound calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (40)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (41)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1-\phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (42)$$

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives only the leading balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (43)$$

Thus $d_{\text{head}}/\sqrt{L}$ is an assumption-bound scaling factor, while the coefficient remains utility-, base-, and convention-dependent. The reported experiments use the exact Cosh warp, including where small- τ truncation is not accurate; Eq. (43) is not a behavioural selector, a support-invariance statement, or a derivation of $c=1$.

Finite tested receipt. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entripoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. These finite points support a fallible zero-search operating prior relative to the tested Geo baseline. They do not establish a continuous basin, basin bounds, interpolation guarantees, near-optimality, or a global optimum.

A.10 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (44)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (45)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (46)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (47)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoff, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (48)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons, $K = 2$ leaves no interior entries and the geometric and Cosh grids coincide.

Large- τ regime. At large τ , $m_\tau = \tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. C and E. This section records the scratch-model and cross-modal experiments that remain supporting evidence. The learned-frequency comparator appears in App. D.

B.2 REPRODUCIBILITY SNAPSHOT

The evidence ladder separates causal identification from architecture, scale, adaptation, frozen-checkpoint, and cross-modal consequences. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 8 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

Table 7: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. C.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. C.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. C.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. G.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. B.3/E.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. E.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. E.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. E.1/E.3
Cross-modal scope	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. B.5

Table 8: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 2–4 (750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 10 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames, extending the scope check to bidirectional 3D RoPE.

Table 9: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

Table 10: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 5 separately measures incompatibility after a frozen post-hoc swap.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R \frac{k}{K-1}\right). \quad (49)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 11: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean±SD	Seed 42	Seed 137	Seed 256	Mean±SD
256	+0.033	+0.029	+0.017	+0.026 ± 0.008	+0.033	+0.029	+0.017	+0.026 ± 0.008
512	−0.478	−0.271	−0.094	−0.281 ± 0.192	+0.061	+0.053	+0.067	+0.060 ± 0.007
1,024	−0.205	−0.191	−0.132	−0.176 ± 0.039	+0.182	+0.156	+0.344	+0.227 ± 0.102
2,048	−0.113	−0.181	−0.143	−0.146 ± 0.034	+0.279	+0.399	+0.701	+0.460 ± 0.218

Under *fixed*, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under *target-matched*, FMRoPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Under the tested target-retargeted policy, the ordering of the two tested policies reverses. The two columns therefore establish that the fixed-support result is conditional on the selected range; they do not identify a support-by-allocation interaction law. The *fixed* column isolates allocation when one table serves multiple lengths (§4.1).

C.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit;

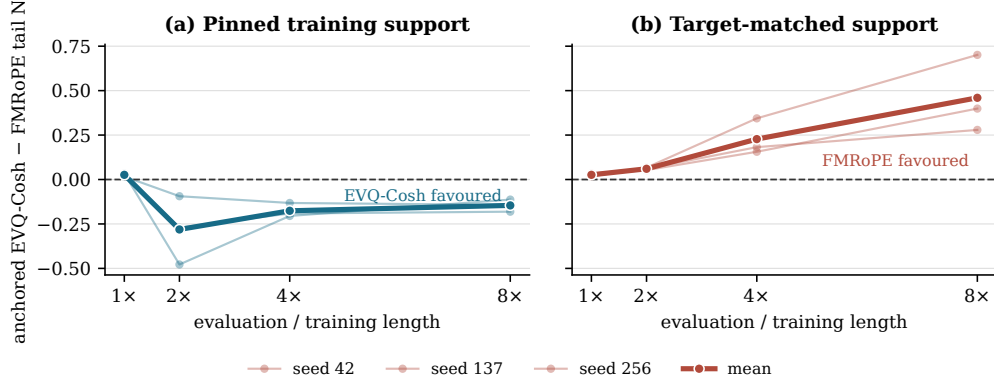


Figure 6: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

Table 12: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

the 32 anchors are paired observations within a seed. This is a two-seed mechanism replication of weight-table co-adaptation, not a replacement for the three-seed exact-range effect and not evidence that either derived table is globally optimal.

C.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the 1.25 $\times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at 0.75 $\times$, 1.00 $\times$ (the formula point), 1.25 $\times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Table 13: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	−0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. The improving direction points inward at both tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream (Merity et al., 2017), a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §4.

D LEARNED-FREQUENCY COMPARATOR

Table 14: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8\text{K}$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §4.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 14 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §4.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). In this protocol, the learned table improves over Geo but remains weaker than EVQ-COSH; the comparison does not identify whether the gap comes from the final table, optimization dynamics, or the positional-parameter learning rate.

Table 15: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

E MATURE-SCALE PROTOCOLS

E.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard $u_k=k/K$ grid with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K *niah_single_1*, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoints are kept separate in the discussion below and in Table 15.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled *Q/K metrics.json* records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 15 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64

at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/+0.0381/-0.0437/-0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

The identity is scientific rather than bitwise trainer pairing. The EVQ-COSH branch uses the reviewed Hugging Face single-GPU loop, whereas the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. Different trainer stacks remain a residual implementation confound, so this is scale persistence from one same-initialisation/same-recipe pair, not a bitwise paired trajectory or a multi-seed causal estimate.

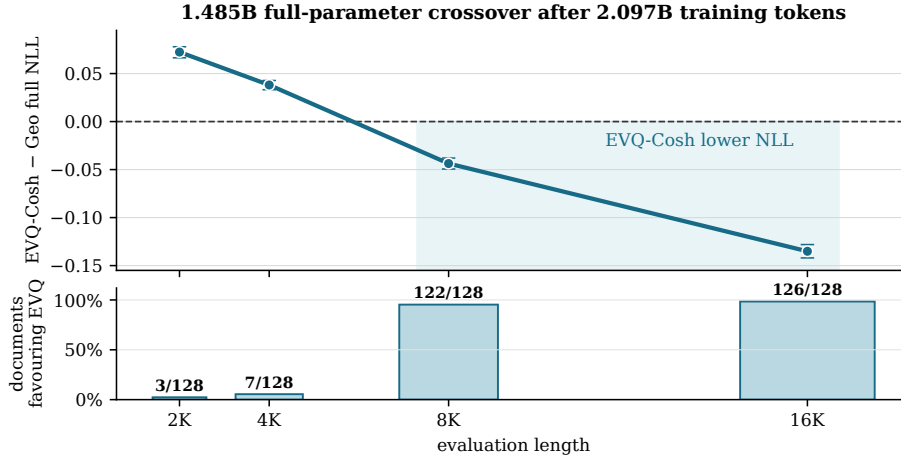


Figure 7: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

E.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 2 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The two reference rows do not: Native retains the pretrained table, while official Transformers YaRN applies its frequency-dependent transport. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $w_k = (1 - \tilde{U}_k)^2$ in the frequency blend of Eq. (10), so less unique pairs move farther.

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalued, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are $[54.88, 64.80]$ points for OLMo and $[0.25, 17.50]$ points for Qwen; the paired

Table 16: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Table 2. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 17 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 17: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The complete OLMo deployment policy is consequently a separate bundled intervention: exact Native is selected when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 18: Compact zero-training deployment endpoints on the frozen OLMo checkpoint. These are complete-policy comparisons, not pure- z effects.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Table 2 gives the corresponding allocation-only comparison.

E.3 LLAMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$;

Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. The contrast is therefore the end-to-end effect of assigning different frequency substrates under a matched adaptation pipeline; it does not isolate pure interior shape because the Native endpoint grid and midpoint-Cosh grid also differ in finite-grid embedding. Under this protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry; it changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 15 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

F FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §5. An amplitude g multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is g^2 . Every table is installed before prefill and retained for the entire request.

F.1 NORMALIZED-INDEX PROFILE

The Qwen-2.5-0.5B-Instruct experiment uses its native $K = 32$ pairs, base $b = 10^6$, and original context length $L_N = 32768$. A reference profile is first evaluated on a 64-pair grid with the same b, L_N :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_N \bar{\omega}_j}{2\pi c_{64}}, \quad (50)$$

$$\bar{m}_j = \text{clip} \left(\frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (51)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of $(j/63, \bar{m}_j)$ at $k/(K-1)$ gives m_k on the target grid. The installed table is $\omega'_k = \omega_k^N 2^{-m_k}$, with fixed amplitude $g = 1 + 0.074 \log 2 = 1.0512928914$. The YaRN-2 comparison uses the Transformers official-equation table and $g = 1 + 0.1 \log 2 = 1.0693147181$; Native uses $g = 1$.

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 19. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval $[2.7627, 9.5835]$; at 32K it is -0.0256 points, with interval $[-3.2630, 3.2179]$.

Table 19: Qwen-0.5B RULER-13 per-task score (%), 20 examples per cell.

Task	Native	32K		YaRN	64K	
		Index			Index	YaRN
Single-1	100.00	100.00	100.00	75.00	100.00	100.00
Single-2	100.00	100.00	100.00	15.00	100.00	75.00
Single-3	100.00	100.00	100.00	30.00	100.00	95.00
MultiKey-1	90.00	95.00	90.00	50.00	90.00	65.00
MultiKey-2	50.00	60.00	35.00	0.00	40.00	25.00
MultiKey-3	5.00	0.00	10.00	0.00	5.00	5.00
MultiValue	62.50	81.25	81.25	22.50	63.75	58.75
MultiQuery	62.50	80.00	77.50	22.50	65.00	50.00
Variable tracking	48.00	42.00	46.00	3.00	28.00	36.00
Common words	2.50	2.00	2.50	7.00	0.50	0.00
Frequent words	46.67	31.67	35.00	31.67	46.67	40.00
QA-1	20.00	10.00	30.00	5.00	15.00	20.00
QA-2	25.00	25.00	20.00	25.00	15.00	20.00

F.2 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let l be the last native frequency pair making more than 32 turns over the reference window and h the first making less than one turn. Write $N = h - l$ and $q = \text{clip}(k - l, 0, N)$. All profiles leave the fast band unchanged and divide the slow-band frequencies by s :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (52)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (53)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (54)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (55)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments $\epsilon_q = m_q - m_{q-1}$ at both band boundaries.

To obtain BM, minimize $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$ with $\epsilon_0 = \epsilon_{N+1} = 0$ and $\sum_{q=1}^N \epsilon_q = 1$. The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (55). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 8 shows the actual profile shapes for the tested transition widths.

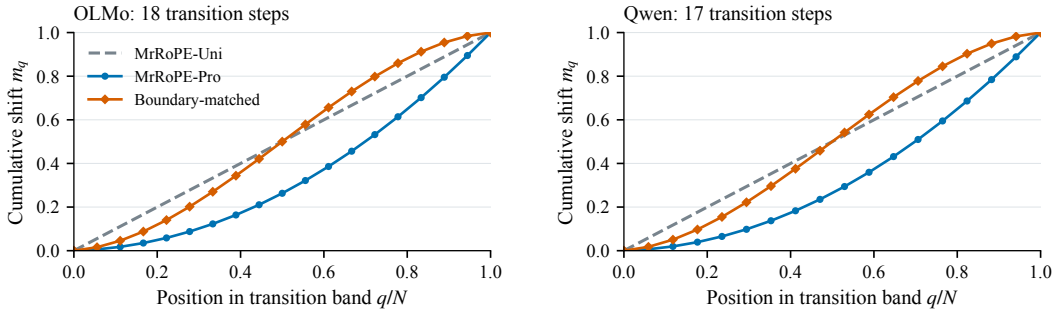


Figure 8: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses $b = 500,000$, reference length 4096, $K = 64$, and $(l, h, N) = (14, 32, 18)$. Qwen-2.5-3B/7B-Instruct use $b = 10^6$, reference length 32768, $K = 64$, and $(l, h, N) = (23, 40, 17)$. The factor-four comparisons use $g = 1 + 0.1 \log 4 = 1.1386294361$ for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

F.3 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909. Its 4K/16K scores are listed in Table 4; MrRoPE-Uni on the same confirmation inputs scores 76.88/32.12%, and official YaRN scores 54.38/6.94%. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 20 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 20: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 4.

Model	Task	Short		Long	
		MrPro	BM	MrPro	BM
OLMo-1B	Single-2	75.00	100.00	0.00	100.00
	MultiKey-2	25.00	100.00	0.00	37.50
	MultiQuery	18.75	87.50	0.00	43.75
	Variable tracking	25.00	70.00	0.00	10.00
	Frequent words	58.33	58.33	16.67	41.67
	QA-1	25.00	75.00	0.00	75.00
Qwen-3B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	75.00	50.00
	MultiQuery	100.00	100.00	93.75	100.00
	Variable tracking	90.00	100.00	75.00	75.00
	Frequent words	83.33	100.00	75.00	75.00
	QA-1	50.00	50.00	50.00	25.00
Qwen-7B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	100.00	100.00
	MultiQuery	100.00	100.00	100.00	100.00
	Variable tracking	100.00	80.00	90.00	60.00
	Frequent words	100.00	100.00	66.67	66.67
	QA-1	0.00	0.00	50.00	0.00

F.4 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 21 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [−3.5618, 7.4742].

The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

Table 21: Natural QA under static $s = 4$. Token F1 uses the whole generated response; EOS counts are recorded separately.

Input length	Task	n	MrPro F1 (%)	BM F1 (%)	MrPro EOS	BM EOS
4K–16K	HotpotQA	166	28.63	33.54	160	163
	2WikiMQA	173	21.55	25.43	169	171
	Qasper	119	13.94	19.06	119	119
	NarrativeQA	61	9.51	11.48	61	61
	MultiFieldQA	112	34.49	37.72	72	72
$\leq 4K$	HotpotQA	5	58.33	51.67	5	5
	2WikiMQA	26	34.62	44.98	26	26
	Qasper	78	26.16	28.14	78	78
	MultiFieldQA	38	50.94	53.27	35	33

G SUPPORTING RESULTS

G.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified empirical $\tau=1.414$ setting. Because the compressed rotary subspace does not instantiate the full-RoPE $d_{\text{rot}}=d_{\text{head}}$ convention in Eq. (39), this experiment tests frequency allocation at fixed architecture rather than the operating rule.

The tracked aggregate is backed by the exact recovered three-seed evaluator JSON and retains every seedwise row. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree. Original checkpoint and training-data artifacts are not packaged, so this row establishes the reported system ordering, not checkpoint-level reproduction or a theorem fixing τ from d_{rope} .

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 22: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this architecture and protocol; the multiplier overlay is the run-specific MLA wavelength-blend operator, not the paper’s pure- z control or an exact reproduction of another scaling method.